6

Topological characteristics in GNN-based recommendation

In recommender systems, GNN-based models learn user preference by leveraging patterns in the user-item bipartite graph. In this section, we investigate whether, and to what extent, these models implicitly or explicitly incorporate the *topological* properties introduced before. To illustrate this, we consider the selected GNN-based recommenders for this monograph: NGCF (Wang *et al.*, 2019), DGCF (Wang *et al.*, 2020), LightGCN (He *et al.*, 2020), SGL (Wu *et al.*, 2021), UltraGCN (Mao *et al.*, 2021), GFCF (Shen *et al.*, 2021), SVD-GCN (Peng *et al.*, 2022b), SimGCL (Yu *et al.*, 2022), LightGCL (Cai *et al.*, 2023), GraphAU (Yang *et al.*, 2023), and XSimGCL (Yu *et al.*, 2024). Specifically, we reinterpret their learning procedures through the lens of the identified *topological* properties, explicitly highlighting their influence.

6.1 NGCF (Wang *et al.*, 2019)

Similar to GCN, NGCF adopts feature transformation and non-linearities within the message passing algorithm. For each ego node, the neighborhood nodes' contribution is rescaled according to the corresponding value in the normalized adjacency matrix. This property helps reduce

disparities between nodes with high and low *node degrees*:

$$\begin{aligned}\mathbf{E}_u^{(l)} &= \text{LeakyReLU}\left(\sum_{i\in\mathcal{N}_u^{(1)}} \mathbf{W}_{\text{neigh}}^{(l)} \tilde{A}_{ui}\mathbf{E}_i^{(l-1)}\right) \\ &= \text{LeakyReLU}\left(\sum_{i\in\mathcal{N}_u^{(1)}} \mathbf{W}_{\text{neigh}}^{(l)} \frac{A_{ui}}{\sqrt{\sigma_u}\sqrt{\sigma_i}}\mathbf{E}_i^{(l-1)}\right),\end{aligned} \tag{6.1}$$

where, for the sake of simplicity, we removed the interdependence factor as it does not encode any topological property. It is worth noticing that $A_{ui} = 1,\ \forall i \in \mathcal{N}_u^{(1)}$, meaning that the denominator solely determines the contribution weighting.

6.2 DGCF (Wang *et al.*, 2020)

This model assumes that user-item interactions are mediated by underlying intents, meaning that user and item representations are learned differently for each specific intent t. Consequently, the adjacency matrix $A_{ui}[t]$ and its normalization are intent-dependent. We reformulate the node embedding as follows:

$$\mathbf{E}_u^{(l)}[t] = \sum_{i\in\mathcal{N}_u^{(1)}[t]} \tilde{A}_{ui}[t]\mathbf{E}_i^{(l-1)}[t] = \sum_{i\in\mathcal{N}_u^{(1)}} \frac{A_{ui}[t]}{\sqrt{\sigma_u[t]}\sqrt{\sigma_i[t]}}\mathbf{E}_i^{(l-1)}[t], \tag{6.2}$$

where the *node degree* serves as a normalization factor that depends on the specific intent t.

6.3 LightGCN (He *et al.*, 2020)

Here, users and items are represented as the linear aggregation of their neighbors. As in NGCF, each neighbor's contribution is weighted according to the corresponding value in the normalized adjacency matrix. We reformulate it as follows:

$$\mathbf{E}_u^{(l)} = \sum_{i\in\mathcal{N}_u^{(1)}} \tilde{A}_{ui}\mathbf{E}_i^{(l-1)} = \sum_{i\in\mathcal{N}_u^{(1)}} \frac{A_{ui}}{\sqrt{\sigma_u}\sqrt{\sigma_i}}\mathbf{E}_i^{(l-1)}. \tag{6.3}$$

Again, the *node degrees* of the ego and neighborhood nodes contribute to weight the aggregated message.

6.4 SGL (Wu *et al.*, 2021)

In SGL, the authors propose to perform graph augmentations (such as node dropout and edge dropout) to obtain two views of the original user-item graph, and ultimately learn two views of each node embedding through the message passing algorithm. In particular, SGL leverages the same message passing as LightGCN, but on the augmented graph views:

$$\begin{aligned}\mathbf{E}_u^{(l)}[t_*] &= \sum_{i \in \mathcal{N}_u^{(1)}} \tilde{A}_{ui}[t_*]\mathbf{E}_i^{(l-1)}[t_*] \\ &= \sum_{i \in \mathcal{N}_u^{(1)}} \frac{A_{ui}[t_*]}{\sqrt{\sigma_u[t_*]}\sqrt{\sigma_i[t_*]}}\mathbf{E}_i^{(l-1)}[t_*],\end{aligned} \tag{6.4}$$

where we compute the normalized version of the adjacency matrix in the specific graph view t_*. The *node degree* information appears within the augmented views of the graph.

6.5 UltraGCN (Mao *et al.*, 2021)

UltraGCN recognizes three major limitations in GCN-based message-passing for collaborative filtering, namely, (i) the asymmetric weight assignment to connected nodes when considering user-user and item-item relationships; (ii) the impossibility to diversify the importance of each type of relation (i.e., user-item, user-user, item-item) during the message-passing; (iii) the over-smoothing effect when stacking more than 3 layers. To tackle such issues, the authors propose to go beyond the traditional concept of explicit message-passing, and approximate the infinite-layer message-passing through the following:

$$\hat{\mathbf{E}}_u = \sum_{i \in \mathcal{N}_u^{(1)}} \frac{A_{ui}\sqrt{\sigma_u + 1}\,\mathbf{E}_i}{\sigma_u\sqrt{\sigma_i + 1}}, \quad \hat{\mathbf{E}}_i = \sum_{u \in \mathcal{N}_i^{(1)}} \frac{A_{iu}\sqrt{\sigma_i + 1}\,\mathbf{E}_u}{\sigma_i\sqrt{\sigma_u + 1}}. \tag{6.5}$$

Note that the procedure is not repeated for layers $l > 1$, as the method surpasses the concept of iterative message-passing. During the optimization, the model first minimizes a constraint loss that adopts

negative sampling to limit the over-smoothing effect:

$$\begin{aligned}\mathcal{L}_{\text{CL}} = &- \sum_{(u,i)\sim\mathcal{T}_+} \frac{1}{\sigma_u}\frac{\sqrt{\sigma_u+1}}{\sqrt{\sigma_i+1}}\ln(\text{Sigmoid}(\hat{\mathbf{E}}_u^\top \hat{\mathbf{E}}_i)) + \\ &- \sum_{(u,j)\sim\mathcal{T}_-} \frac{1}{\sigma_u}\frac{\sqrt{\sigma_u+1}}{\sqrt{\sigma_j+1}}\ln(\text{Sigmoid}(-\hat{\mathbf{E}}_u^\top \hat{\mathbf{E}}_j)),\end{aligned} \tag{6.6}$$

where $\mathcal{T}_+ = \{(u,i) \mid R_{ui} = 1\}$ and $\mathcal{T}_- = \{(u,j) \mid R_{uj} = 0\}$, while $\ln(\cdot)$ and $\text{Sigmoid}(\cdot)$ are the logarithm and sigmoid function. Then, they also take into account the item-item projected graph $\mathcal{G}_\mathcal{I}$ and minimize:

$$\mathcal{L}_{\text{IL}} = - \sum_{(u,i)\sim\mathcal{T}_+} \sum_{j\in\text{topK}(\mathbf{R}^{\mathcal{I}\mathcal{I}}_{i*})} \frac{R^{\mathcal{I}\mathcal{I}}_{ij}}{\sigma^{\mathcal{I}\mathcal{I}}_i - R^{\mathcal{I}\mathcal{I}}_{ii}} \sqrt{\frac{\sigma^{\mathcal{I}\mathcal{I}}_i}{\sigma^{\mathcal{I}\mathcal{I}}_j}} \ln(\text{Sigmoid}(\hat{\mathbf{E}}_u^\top \hat{\mathbf{E}}_j)), \tag{6.7}$$

where $\text{topK}(\cdot)$ retrieves the top K values of a matrix row-wise, and $\sigma^{\mathcal{I}\mathcal{I}}_*$ is the node degree calculated on $\mathcal{G}_\mathcal{I} = \{\mathcal{I}, \mathbf{A}^{\mathcal{I}\mathcal{I}} = \mathbf{R}^{\mathcal{I}\mathcal{I}}\}$. Thus, the model formulation encodes the *node degrees* information in both the user-item and the *item-item* co-occurrence graphs.

6.6 GFCF (Shen *et al.*, 2021)

The authors of GFCF re-interpret all main GNN-based recommender systems under the lens of graph signal processing. Eventually, they design a strong closed-form algorithm to predict novel user-item interactions:

$$\hat{\mathbf{R}}_{u*} = \mathbf{R}_{u*}\left(\left(\tilde{\mathbf{A}}^{\mathcal{U}\mathcal{I}}\right)^\top \tilde{\mathbf{A}}^{\mathcal{I}\mathcal{U}} + t\left(\mathbf{D}^{\mathcal{I}\mathcal{I}}\right)^{-\frac{1}{2}} \text{topK}(\mathbf{U})\text{topK}(\mathbf{U})^\top \left(\mathbf{D}^{\mathcal{I}\mathcal{I}}\right)^{+\frac{1}{2}}\right), \tag{6.8}$$

where the superscripts $\mathcal{U}\mathcal{I}$, $\mathcal{I}\mathcal{U}$, and $\mathcal{I}\mathcal{I}$ stand for the user-item, item-user, and item-item partitions of the adjacency matrix, respectively. As $\tilde{\mathbf{A}}^{\mathcal{U}\mathcal{I}} = \left(\mathbf{D}^{\mathcal{U}\mathcal{I}}\right)^{-\frac{1}{2}} \mathbf{A}^{\mathcal{U}\mathcal{I}} \left(\mathbf{D}^{\mathcal{U}\mathcal{I}}\right)^{-\frac{1}{2}}$ and $\tilde{\mathbf{A}}^{\mathcal{I}\mathcal{U}} = \left(\mathbf{D}^{\mathcal{I}\mathcal{U}}\right)^{-\frac{1}{2}} \mathbf{A}^{\mathcal{I}\mathcal{U}} \left(\mathbf{D}^{\mathcal{I}\mathcal{U}}\right)^{-\frac{1}{2}}$, we explicitly highlight the presence of the *node degrees* information of the user-item graph and *item-item* co-occurrence graph within the GFCF formulation.

6.7 SVD-GCN (Peng *et al.*, 2022b)

By reinterpreting the message-passing mechanism introduced in LightGCN, SVD-GCN leverages both even-hop (user-user, item-item) and odd-hop (user-item, item-user) message aggregations, along with the singular vectors and values of the normalized adjacency matrix $\tilde{\mathbf{A}}^{\mathcal{UI}} = \left(\mathbf{D}^{\mathcal{UI}}\right)^{-\frac{1}{2}} \mathbf{A}^{\mathcal{UI}} \left(\mathbf{D}^{\mathcal{UI}}\right)^{-\frac{1}{2}}$. The user (item) embeddings are computed as follows:

$$\hat{\mathbf{E}}_u = (\mathbf{U})\exp(t_1\mathbf{V})\mathbf{W}, \tag{6.9}$$

where $\mathbf{U}$ contains the singular vectors of $\tilde{\mathbf{A}}^{\mathcal{UI}}$, t_1 is a tunable parameter, $\mathbf{V}$ is vector containing the largest singular values on $\tilde{\mathbf{A}}^{\mathcal{UI}}$, and $\mathbf{W}$ is a trainable weight matrix. A connection exists between the largest singular value $v_{\max} \in \mathbf{V}$ and the *maximum node degree* $\max(\mathcal{D})$ in the user-item matrix, captured by the following inequality:

$$v_{\max} \leq \frac{\max(\mathcal{D})}{\max(\mathcal{D}) + t_2}, \tag{6.10}$$

where t_2 is an additional tunable hyperparameter controlling the spacing among singular values. Moreover, to capture *even-hop connections* in the bipartite graph, the loss function incorporates an additional component that depends on user-user (item-item) interactions. Specifically, this component considers *users (items) co-occurrences* in the graph as follows:

$$\mathcal{L}_{\text{UL}} = \sum_{u \in \mathcal{U}} \sum_{(u,v) \sim \mathcal{T}_+^{\mathcal{UU}},\, (u,w) \sim \mathcal{T}_-^{\mathcal{UU}}} \ln \text{Sigmoid}(\hat{\mathbf{E}}_u^\top \hat{\mathbf{E}}_v - \hat{\mathbf{E}}_u^\top \hat{\mathbf{E}}_w), \tag{6.11}$$

where $\mathcal{T}_+^{\mathcal{UU}} = \{(u,v) \mid A_{uv}^{\mathcal{UU}} \neq 0\}$ and $\mathcal{T}_-^{\mathcal{UU}} = \{(u,w) \mid A_{uw}^{\mathcal{UU}} = 0\}$, while $\text{Sigmoid}(\cdot)$ is the sigmoid function.

6.8 SimGCL (Yu *et al.*, 2022)

SimGCL surpasses the traditional concept of graph contrastive learning by applying augmentations on the node embeddings rather than on the graph structure. Thus, from a topological perspective, the presence of the *node degree* information is equal to that of other models such as

NGCF and LightGCN:

$$\begin{aligned}\mathbf{E}_u^{(l)}[t_*] &= \sum_{i \in \mathcal{N}_u^{(1)}} \tilde{A}_{ui} \mathbf{E}_i^{(l-1)}[t_*] + \Delta^{(l)}[t_*] = \\ &= \sum_{i \in \mathcal{N}_u^{(1)}} \frac{A_{ui}}{\sqrt{\sigma_u}\sqrt{\sigma_i}} \mathbf{E}_i^{(l-1)}[t_*] + \Delta^{(l)}[t_*],\end{aligned} \tag{6.12}$$

where t_* stands for one of the two computed views.

6.9 LightGCL (Cai *et al.*, 2023)

LightGCL proposes a simple yet effective graph contrastive learning solution, where two views of the user and item embeddings (obtained from a global and local perspective on the user-item graph) are contrasted. Regarding the global view, it is obtained through the same message passing formulation proposed in (Xia *et al.*, 2022a):

$$\mathbf{E}_u^{(l)} = \text{LeakyReLU}\left(\text{drop}(\tilde{\mathbf{A}}_{u*})\mathbf{E}_i^{(l-1)}\right) + \mathbf{E}_i^{(l-1)}, \tag{6.13}$$

where the model applies the edge dropout function to the normalized adjacency matrix $\tilde{\mathbf{A}} = \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-12}$. Thus, as in other described models, the *node degree* component is explicitly highlighted in the formulation.

6.10 GraphAU (Yang *et al.*, 2023)

In a similar manner as (Wang *et al.*, 2022a), the authors redefine the uniformity and alignment paradigm to graph-based recommendation within this model. From a topological viewpoint, GraphAU shows the same contribution as LightGCN, by leveraging the normalized adjacency matrix where the *node degree* is explicitly encoded:

$$\mathbf{E}_u^{(l)} = \sum_{i \in \mathcal{N}_u^{(1)}} \tilde{A}_{ui} \mathbf{E}_i^{(l-1)} = \sum_{i \in \mathcal{N}_u^{(1)}} \frac{A_{ui}}{\sqrt{\sigma_u}\sqrt{\sigma_i}} \mathbf{E}_i^{(l-1)}. \tag{6.14}$$

6.11 XSimGCL (Yu *et al.*, 2024)

Unlike other contrastive models, XSimGCL leverages the same contrasted view of node embeddings for both the recommendation task and

the contrastive learning task; it then applies a cross-layer contrastive module, which calculates the InfoNCE between the same node view calculated at different propagation layers. From a topological perspective, XSimGCL follows the same message passing as SimGCL, where the normalized adjacency matrix encodes the *node degree* information:

$$\begin{aligned}\mathbf{E}_u^{(l)}[t] &= \sum_{i \in \mathcal{N}_u^{(1)}} \tilde{A}_{ui} \mathbf{E}_i^{(l-1)}[t] + \Delta^{(l)}[t] = \\ &= \sum_{i \in \mathcal{N}_u^{(1)}} \frac{A_{ui}}{\sqrt{\sigma_u}\sqrt{\sigma_i}} \mathbf{E}_i^{(l-1)}[t] + \Delta^{(l)}[t],\end{aligned} \tag{6.15}$$

where, this time, only one view is considered and calculated.

6.12 Topology and GNN models: an overview

In Table 6.1, we summarize the main topological contributions of each selected GNN model.

Overall, the most evident trend is the explicit encoding of the *node degree* information within the models' formulations, mainly adopted in the calculation of the normalized adjacency matrix of the user-item and (in one case) the item-item graph. Additionally, some GNN models work on the graph structure of the user-item graph by modifying it or weighting the importance of its different contributions. Finally, a few models do not limit their perspective to the user-item graph, but propose leveraging the user-user and item-item projections to reveal hidden node relationships, which could become fundamental in recommendation.

Final remarks

The analysis reveals that GNN-based recommenders explicitly incorporate node degrees in their formulations, while other topological properties, such as clustering coefficient and assortativity, may only be implicitly captured—if at all. Therefore, our explanatory framework, detailed in Section 7, also serves to investigate which topological characteristics GNN-based recommender systems might unintentionally or implicitly learn during their pipeline.

Table 6.1: Selected GNN models for this monograph and how they handle topological properties according to their implemented strategies for the the graph structure, the prediction, and the loss function.

Model	Graph structure	Prediction	Loss function
NGCF	✗	Node degree in the normalized adjacency matrix	✗
DGCF	Separate graph structure for each intent	Node degree in the normalized adjacency matrix	✗
LightGCN	✗	Node degree in the normalized adjacency matrix	✗
SGL	Node dropout and edge dropout applied to obtain different graph views	Node degree in the normalized adjacency matrix	✗
UltraGCN	✗	Node degree for normalizing the adjacency matrix in a custom way	The item-item co-occurrences are used for the IL component
GFCF	✗	Node degree in the normalized adjacency matrix and item-item degree matrix	✗
SVD-GCN	✗	Node degree in the normalized adjacency matrix and connection between max eigenvalue and max node degree	The user-user and item-item co-occurrences are used for the UL/IL components
SimGCL	✗	Node degree in the normalized adjacency matrix	✗
LightGCL	Edge dropout applied to address overfitting	Node degree in the normalized adjacency matrix	✗
GraphAU	✗	Node degree in the normalized adjacency matrix	✗
XSimGCL	✗	Node degree in the normalized adjacency matrix	✗